



The Effects on Driving Behavior When Using a Head-mounted Display in a Dynamic Driving Simulator

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Driving simulators are established tools used during automotive development and research. Most simulators use either monitors or projectors as their primary display system. However, the emergence of a new generation of head-mounted displays has triggered interest in using these as the primary display type. The general benefits and drawbacks of head-mounted displays are well researched, but their effect on driving behavior in a simulator has not been sufficiently quantified.

This article presents a study of driving behavior differences between projector-based graphics and head-mounted display in a large dynamic driving simulator. This study has selected five specific driving maneuvers suspected of affecting driving behavior differently depending on the choice of display technology. Some of these maneuvers were chosen to reveal changes in lateral and longitudinal driving behavior. Others were picked for their ability to highlight the benefits and drawbacks of head-mounted displays in a driving context.

The results show minor changes in lateral and longitudinal driver behavior changes when comparing projectors and a head-mounted display. The most noticeable difference in favor of projectors was seen when the display resolution is critical to the driving task. The choice of display type did not affect simulator sickness nor the realism rated by the subjects.

CCS Concepts: • **Computing methodologies** → **Virtual reality**; **Perception**; • **Applied computing** → *Transportation*;

Additional Key Words and Phrases: Head-mounted display, driving simulator, driver behavior

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1 INTRODUCTION

Driving simulators as a tool during automotive development and research have several benefits compared to using an actual vehicle. The simulated environment can be kept consistent between test runs, which allows for refined tests where only the explanatory variables are varied. Driving simulators also allow for scenarios that are too dangerous to perform in a real environment or scenarios that involve complex interactions between actors that would be almost impossible to orchestrate in real life.

A challenge with driving simulators is to design them so that they excite a realistic driving behavior. There are simulators at many different levels of fidelity, ranging from small desktop-based simulators that use a steering

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wheel and pedals intended for computer games to large simulators, which can accommodate actual vehicles inside 360° projection domes articulated by large 6-DOF motion systems.

Another challenge with driving simulators is to limit the amount of simulator sickness induced by the simulator. Simulator sickness can occur when there is conflicting sensory input from the visual and vestibular systems [28]. This conflict can originate from delays in the simulator's imaging system, as well as from the inability of the simulator motion system to reproduce the correct motion feedback. When starting to feel the onset of simulator sickness, the driver may involuntarily adapt their driving behavior to alleviate the symptoms. This adaptation can lead to questionable experimental validity. Previous research has shown that display systems' performance affects the level of simulator sickness [11].

The most common display system for small simulators uses one or more monitors placed in front of the driver. Monitors are very cost effective, although it is impossible to cover a large field-of-view without interruption by the screen bezels. If an uninterrupted field-of-view is desired, then the standard solution is to utilize projector-based solutions where multiple projectors are combined to give a large, high-resolution view. The most modern systems reach resolutions that are getting close to the limits of the performance of the human eye [31].

The most common use of virtual reality in the automotive sector has been as a tool for design evaluation [3]. In this context, it is essential to have high-resolution images to make the correct design decisions. Thus the most common display solution has been to use CAVE solutions [17], since they can provide higher resolutions than the current generation of head-mounted displays.

Head-mounted displays have not been commonly used, since they have a history of suffering from low resolution, low field-of-view, high tracking latency, and high price. The focus of the image inside a head-mounted display is at a fixed distance, which may result in *Vergence-Accommodation Conflict* [19]. In turn, this conflict can result in headaches and fatigue for the user [20].

During recent years, the technological development of head-mounted displays has surged, with improved tracking performance, higher resolution displays, and wider field-of-view. As a result, the question of replacing computer screens and projectors as the primary solution for presenting the virtual environment to the users has reappeared. The main appeal with this is the possibility of increased realism and lower cost of investment compared to traditional display solutions.

2 RELATED WORK

An early study by Schill et al. [32] describes how Daimler-Benz evaluated a head-mounted display inside their dynamic driving simulator. They performed a between-group study ($n = 48$), where they compared the difference between projectors and head-mounted display. They compared tasks involving speed perception, lane-change maneuvers, and a car following task. They found similar driving behavior for the performed tasks, and the subjectively rated simulator sickness in the two groups was also similar.

Burns and Saluäär [5] compared real driving with driving in a static simulator equipped with a head-mounted display in a within-group study ($n = 11$). The researchers compared speed perception, time to complete turns, lane-keeping as well as subjective workload. They also recorded the visual behavior via cameras inside the head-mounted display as well as inside the vehicle. Their results showed that the subjects used similar times to complete turns in both studied conditions. However, they had slower speed and worse lane-keeping after turning when they wore the head-mounted display. Their visual behavior was also affected, and they needed longer glance durations before turning compared to real driving. In addition, the subjects rated a higher required mental workload when using the head-mounted display.

Coates et al. [7] integrated a head-mounted display with a wide field-of-view (120° horizontal and 67° vertical) in the Renault dynamic simulator. The authors point out the need for a wide horizontal field-of-view for speed perception and a wide vertical field-of-view to see both the environment and the dashboard at the same time.

A study by Carlozzi et al. [6] investigated the possibility of using driving simulators for training subjects with spinal cord injuries to drive. Their study compared a head-mounted display with a three-monitor setup for a

simple static driving simulator in a between-group study ($n = 54$). Their results show less lateral control for the group using the head-mounted display solution, but they saw no significant difference in motion sickness rate between the two groups.

Grabe et al. [13] compared a projection screen with a head-mounted display without head-tracking. They used a within-group study to test with both 90° and 45° field-of-view while the drivers ($n = 12$) performed a slalom course in a motion-base simulator. The deviation from an ideal path was measured. The field-of-view angle had no significant effect, but the head-mounted display had a greater error for each tested condition. The head-mounted display solution performed worse even when the researcher used the maximum available field-of-view in the head-mounted display, which was greater than the available field-of-view for the projection screen. Their results suggest that the selected slalom task is more dependent on central vision. The authors also saw no significant difference when the motion system of the simulator was disabled. The angular resolution, brightness, and contrast were superior in the head-mounted display compared to the projection screen, suggesting that none of these factors can explain why the drivers performed worse while wearing the head-mounted display.

Aykent et al. [1] performed a within-group study where the subjects ($n = 14$) made a subjective comparison between a head-mounted display and a three-monitor solution in a fixed-base simulator. Their results favored the monitor-based solution due to the low field-of-view and high motion sickness in the head-mounted display, while the subjective level of presence was scored equivalently for both compared setups.

In a similar fixed-base driving simulator study, Colombet et al. [8] compared a head-mounted display to a four-sided CAVE virtual reality room. The subjects ($n = 12 + 12$) were instructed to drive two different scenarios, one with several changes in longitudinal motions (acceleration/deceleration) and one that had several turns creating yaw rotations. The subjects performed the two scenarios at both low and high speeds. The results showed that the head-mounted display users were more susceptible to motion sickness during fast longitudinal motions. For yaw rotational motions, the rate of simulator sickness was similar regardless of the chosen display technology.

Weidner et al. [36] also used a static simulator to compare traditional projectors, active stereo projectors, and a head-mounted display in a between-group study ($n = 94$). The driving task was to perform a lane-change maneuver. The study found no significant effects on measured driving behavior. They saw an increase in simulator sickness with the head-mounted display, although the difference was only significant when comparing the active stereo projector and the head-mounted display.

A recent study by Zöller et al. [37] performed a comparison between projectors and head-mounted displays in a static simulator. The study, which used a within-group design ($n = 41$), also incorporated a mono-camera to the head-mounted display to show the actual interior of the vehicle and the drivers' own hands. They measured the difference in immersion, speed-perception, distance estimation, visual acuity, and reaction time during a simulated rural road scenario. The results showed no significant difference in immersion or distance estimation. The speed-perception task resulted in under-estimation of own speed when driving with a head-mounted display compared to an over-estimation for projectors. The visual acuity test showed that the subjects were able to identify road signs much earlier with projectors compared to the head-mounted display (mean distance 79 m compared to 24 m). The head-mounted display case also showed slower reaction times. During the post-study interview, the subjects stated some physical discomfort of using the head-mounted display, and a majority showed a preference for the projector-based graphics. Hartfiel and Stark [15] investigated if introducing a midsize hexapod motion system might alleviate the motion sickness experienced when driving while wearing a head-mounted display with either the motion system enabled or disabled. The subjects ($n = 100$) drove 4 minutes in one of three scenarios; highway driving, rural driving, or urban driving. The results hint at a small reduction in motion sickness when using a motion system but showed no statistically significant effect.

An alternative to using a motion system is to use a simulated environment inside an actual vehicle on a test track [12]. The main benefit of this setup is that there is no need for any vehicle dynamics simulation, and the motion feedback from the vehicle is correct by default. Using this type of setup, Benz et al. [2] compared the

level of motion sickness and presence between using a monitor, a projector, and a head-mounted display in a within-group study ($n = 60$). The monitor-based setup used a 65-inch widescreen display mounted flush outside the windshield. The projector was mounted in the ceiling and displayed its image on the inside of the windshield. The projector setup allowed a larger field-of-view compared to the monitor solution. The image viewpoint in the monitor setup and the projector setup was not oriented to the driver's dynamic head position. In contrast, the head-mounted display used an optical tracker to determine the dynamic orientation and position of the head. The results showed a significant decrease in motion sickness with the monitor-based setup compared to the head-mounted display setup. The projector-based setup indicated a decrease in motion sickness as well. The subjects rated the presence highest for the monitor-based setup followed by the head-mounted display setup, but that effect was not statistically significant.

3 FOCUS OF CURRENT RESEARCH

If head-mounted displays are to be used in driving simulator applications, then there is a need to know if the technology in itself affects driving behavior. Previous research has mainly focused on general driving either in highway, rural, or urban environments. This study focused on finding more specific answers to why different display types may elicit particular responses in the subjects, hence targeting the study toward specific driving tasks. Thus, the goal was to find maneuvers where the response was different depending on the display type and find which maneuvers are unaffected.

Previous meta-analysis [4] have indicated problems with head-mounted displays in scenarios that require any of the following conditions:

Fast side-to-side head motions. Users of head-mounted displays are susceptible to latency when doing rapid head turns [16]. The reason being that any latency will introduce a phase error in the orientational tracking, which becomes particularly noticeable when the direction of the head-movement changes.

Significant lateral motion of the vehicle. Any swift side-to-side movement of the vehicle performed by the driver will cause the entire scene to move across the retina in a similar way to head rotations, thereby evoking the same problematic behavior as above. For simulators, this motion needs to be replicated by the motion system, and any discrepancies between the visual feedback and motion feedback are likely to increase motion sickness.

High visual acuity. Most countries define a legal requirement for corrected visual acuity to be considered fit to drive. In the EU, the minimum requirement is a binocular visual acuity of at least 0.5 [33]. This level of acuity roughly translates into an angular display resolution of 30 pixels/°. This can be compared to the current generation of head-mounted displays, which ranges from 10 to 15 pixels/°.

Interaction with physical hardware inside the vehicle cabin. When putting on a head-mounted display, the user blocks out all but the virtual environment. To be immersed entirely can be beneficial in some situations, but it can be problematic if the user is supposed to interact with real hardware in the real environment. The head-mounted display also blocks out the users' own body, which can be problematic. This lack of self-embodiment forces the users to rely solely on proprioception to determine the position of their own hands and may cause sensory miscues. There are ways around this problem using hand-tracking and replicating the real hardware as objects in the virtual world [22].

There are also potential benefits of using head-mounted displays. Drivers in simulators tend to use longer safety distances compared to the real-world due to a reduction in speed perception. Speed perception may improve from having motion parallax due to the dynamic changes of the viewpoint caused by the changes in the driver's head position [26].

The dynamic viewpoint combined with the binocular display technology enables more depth cues in the near to medium range (*below 30 m*) in a head-mounted display compared to projectors or monitors [29]. These added depth cues may enable the driver to make better estimates of the position and orientation of their vehicle.

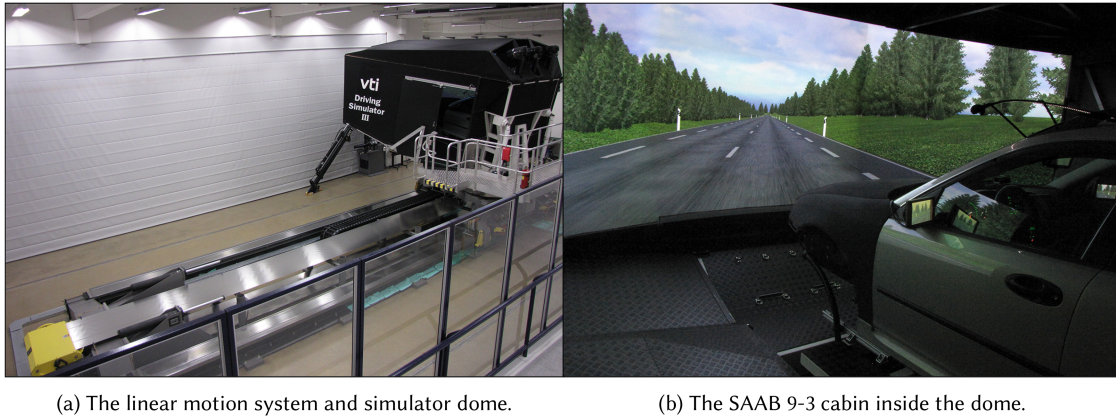


Fig. 1. The VTI Simulator 3.

4 METHOD

The study presented in this article used a within-group design with two conditions; One with projector-based graphics and one using a head-mounted display. The visual display type conditions were balanced between the subjects, where half of the subjects started driving the task with projector-based graphics, and the other half started with a head-mounted display. The balanced design was used to reveal any order effects between the studied conditions, which was especially important given that the amount of simulator sickness may accumulate from one condition to the next.

4.1 Apparatus

The study utilized the VTI Simulator 3 [25], which has a motion system with 4 degrees of freedom. The foundation of this motion system is a linear system spanning 7.5 m (see Figure 1(a)). This linear system can achieve accelerations up to 8.0 m/s^2 and provide realistic lateral acceleration feedback for high dynamic maneuvers. On top of the linear system is the projection dome, which can be articulated with hydraulics to provide tilt coordination, i.e., pitch and roll of the dome. Tilt coordination can be used to provide simulated accelerations in both lateral and longitudinal directions. The vibration table inside the dome offers 4 degrees of freedom, i.e., heave, surge, roll, and pitch. This vibration table is used to give realistic high-frequency feedback representing the road surface contact and suspension forces. The front half of a real SAAB 9–3 cabin is attached to the vibration table (see Figure 1(b)).

The virtual test track used in this study consists of two straight segments connected by two hairpin curves. The primary motivation of using a test track designed in this fashion was to facilitate a future repeat study of the same maneuvers but on a real test track. Consequently, the dimensions had to be kept within the available road surface at the intended proving ground. The two hairpin curves were selected so that their radius matched the smallest roundabouts recommended in Sweden. However, this road design had consequences for the motion cueing strategy in the simulator.

The simulator employs a motion cueing strategy that uses a linear motion system to match the lateral lane position. This strategy results in realistic side-to-side acceleration for momentary maneuvers. For longitudinal acceleration and sustained lateral acceleration, the simulator uses pitch and roll motion. The narrow radius of the hairpin curves exceeded the valid range of the motion cueing algorithm, which resulted in incorrect roll motion in the simulator. Since the remaining parts of the virtual roads contain few parts that use the roll motion, the decision was made to disable the rolling motion. Regrettably this meant disabling the pitch motion as well. Hence the motion feedback relied solely on the vibration table and the lateral linear system.

4.1.1 Projector System. The projector system uses six SONY HW30 SXRD-projectors with FullHD resolution. The projectors are positioned in a 3-by-2 configuration with a combined horizontal field-of-view of 115° seen from the driver's seat. This results in average angular resolution of 42.7 pixels/°. The image update frequency for the projectors is 60 Hz.

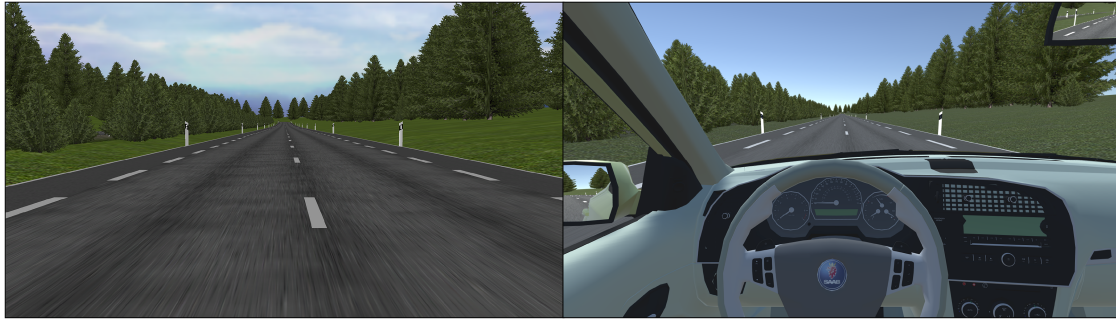
4.1.2 Head-Mounted Display. There are particular factors to consider when deciding which head-mounted display to use inside a moving vehicle. The head-mounted display's weight must be kept as low as possible, as a heavy device can cause neck-strain during high dynamic maneuvers. A very bulky device may also limit the user's motion as such a device will collide with the cabin interior. The available tracking system must also work inside the relatively small vehicle cabin, either with the device's built-in tracking system or the ability to integrate with a suitable third-party tracker.

The head-mounted display selected for the study was an Oculus Rift CV1, which has an angular resolution of 13.8 pixels/° (0.23 pixels/arcmin [21]) in the center of the visual field. This angular resolution was well in line with other head-mounted displays available on the open market when the study was performed [23]. The image update frequency for the Oculus Rift is 90 Hz, which was among the highest available at the time of the study [30]. The field-of-view inside the head-mounted display is 94° horizontal and 93° vertical. There were devices with wider field-of-view available on the market, but these devices were bulkier and had higher weight, lower angular resolution, and lower update frequency [24].

The Oculus Rift CV1 uses an inertial-optical tracking system. The inertial part measures the linear and rotational accelerations of the headset. These measurements are integrated twice to calculate the current position and orientation. The drawback of this integration is that any noise in the sampled measurements will accumulate and cause drift in position and orientation. Consequently, there is a need for an optical part (which Oculus has named the *constellation tracking system*), which uses two or more fixed cameras placed around the user. The optical system uses computationally expensive image processing, but it can operate at a lower update rate, since its primary purpose is to correct for drift caused by the inertial system. However, the Oculus Rift CV1 is not designed for use on a dynamic platform, and any motion of the dynamic platform gets incorrectly interpreted by the inertial system as head movements instead. The optical tracking system could, in theory, correct for this, but that would require placing the tracking cameras outside of the moving platform. This camera placement is impossible in this case, since the simulator is covered and has a much broader envelope than the conceivable tracking area for the Oculus tracking cameras. In theory, it should be possible to subtract the motion information from the simulator motion system to compensate for its contribution to the inertial information inside the head-mounted display [10]. Regrettably, the motion tracking algorithms of the Oculus Rift are closed source, which makes this type of compensation hard to implement. Consequently, the built-in tracking system was disabled, replacing it with a third-party tracker, in this case the OptiTrack V120, which uses two cameras operating at 120 Hz and tracks the position of a group of reflective markers placed on the object to ascertain its position and orientation.

4.1.3 Virtual Environment. To perform the study, there was a need for a virtual environment that featured a test track whose geometry allowed for the desired maneuvers. This track has the following layout; The starting point is a straight road with two lanes each 3.5 m wide (see Figure 2). Outside these lanes, there is a 1.5 m verge. The straight part is 680 m long and has a decline of 0.5 %. After the straight road section, the right lane narrows over 100 m until only the left lane remains. The road then makes a 180° turn with an outer radius of 11 m. After this turn, the road widens over 100 m to two lanes again. A new straight section of 680 m follows, which has an incline of 0.5%. The straight section is followed by yet another narrowing section and a 180° turn, which finally widens and reconnects with the starting point.

The road was built by hand using the OpenDrive logical description and then converted to a visual environment using in-house image generation software tools. The visual environment was exported from this tool,



(a) The projector-based application.

(b) The virtual reality application.

Fig. 2. The virtual environments as seen from within the two visual display types.

converted, and imported into a custom Unity3D application. This Unity3D application used the positions and orientations from the optical tracker to control the head-mounted display. In this application, a virtual vehicle interior was added to replicate the actual vehicle interior the subject was sitting in (see Figure 2(b)). The vehicle interior replicated the steering wheel behavior and pedal movement of the simulator cabin. The speedometer and tachometer were also readable inside the virtual cabin and mirrored the actual gauges' behavior in the simulator cabin. No tracking of the driver's hands was featured in this study.

4.2 Driving Tasks

Five different driving tasks were selected to highlight the differences between projector-based display systems and head-mounted displays. Each of the selected tasks is described in detail below, together with each task's evaluation criteria.

4.2.1 Speed Perception. The subjects were asked to drive this task with the speedometer and tachometer disabled to investigate if head-mounted displays improve speed perception. The subjects started from standstill and were asked to accelerate up to what they perceived to be 40 km/h. As soon as they felt they had reached the instructed speed, they had been instructed to press a button on the steering wheel.

When they pressed the button, they were instructed to decrease their speed to 20 km/h. Once again, pressing the button when they felt they had reached the instructed speed.

The task then repeated with 40 and 20 km/h in the same fashion. The driven speed was recorded to analyze the deviation compared to the instructed reference speed. Previous research has shown that increased field-of-view results in better speed perception [9]. The head-mounted display allows the user to turn the head for a more immersive experience. Additional depth cues from the binocular display system, coupled with the motion parallax from head motion, should allow for more depth cues than projection-based graphics [26]. Thus the hypothesis is that the deviation would be smaller for head-mounted displays.

4.2.2 Low-speed Lane Change. The low-speed lane change task involved navigating a double-lane-change maneuver (see Figure 3). The subjects were instructed to drive no faster than 20 km/h and try to keep as centered between the cones as possible. Each subject repeated the low-speed lane-changing maneuver twice before starting the next task.

The average lateral deviation from an ideal centerline path between the cones was calculated using Equation (1), where $r_{ideal}(s)$ denotes the ideal center path between the cones (shown in bold in Figure 3). The actual driven path by each subject is denoted as $r_{test}(s)$. Consequently, the accumulated deviation can be calculated as

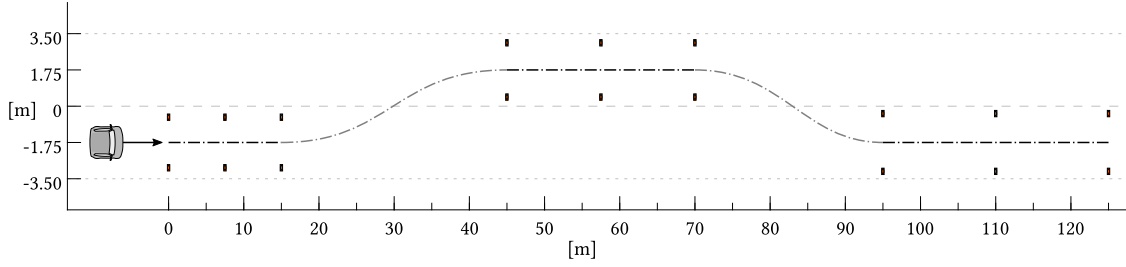


Fig. 3. The double-lane-change maneuver used in both the low-speed and the medium-speed task.

the sum of the absolute difference between the actual and ideal path over the driven distance (d). Dividing the accumulated deviation with the driven distance will result in the average deviation.

In addition to lateral deviation, the number of steering wheel reversals needed to perform each maneuver was measured. A steering wheel reversal was defined as a change in the direction of the steering wheel's rotation. All reversals that involved rotations less than 0.5° were removed to filter out measurement noise.

The hypothesis is that this low-speed lane change would benefit from the added depth cues introduced by the head-mounted display, which would result in lower average deviation and fewer steering wheel reversals.

$$\text{Average Deviation} = \frac{1}{d} \int_0^d |r_{\text{ideal}}(s) - r_{\text{test}}(s)| ds. \quad (1)$$

4.2.3 Medium Speed Lane Change. The medium speed lane change task used the same type of lane change setup, the difference being that the subjects were instructed to drive in 40 km/h. The idea was to force the subject to make fast lateral movements with the vehicle, which previous research indicated could be hard to do using a head-mounted display.

The task used the same measurements as the low-speed lane change task. Each subject repeated the medium speed lane-changing maneuver twice before starting the next task. In this case, the hypothesis is that driving with a head-mounted display would have more pronounced deviations and more steering wheel reversals compared to driving with projector-based graphics.

4.2.4 Sign Detection. The subject performed a sign detection task to examine the effects of the reduced visual acuity in head-mounted displays. The task started with the subjects positioned with the vehicle positioned on the centerline of the road. A road sign appeared on the centerline of the road far away in front of the vehicle. This road sign could have one of two possible symbols (see Figure 4), selected randomly. The subjects were instructed to drive toward the road sign in 40 km/h. As soon as they could distinctly identify the symbol on the sign, their task was to steer and pass on the side indicated by the sign. The subjects performed four of these maneuvers per display type.

An individual with perfect vision should be able to discern objects extending 1 arcmin [35], which corresponds to a width of the identifiable features of 0.58 mm at 2-m distance. The selected road sign is designed to be used for testing visual acuity in a vehicle driving context. The width of the black bars on the symbol and the white distance between them are 58 mm. Thus, if viewed in real life and assuming perfect vision, then the symbol should not be possible to decipher until being closer than 200 m. The projector system used in this study have a resolution of 42.7 pixels/°. At least 2 pixels are needed to form a distinguishable feature, and thus the narrowest object that a individual with perfect version can discern extends 2.81 arcmin (see Equation (2)). Given the resolution of the projector-based setup used in this study, the theoretical limit is 71.0 m (see Equation (3)). The same calculation



Fig. 4. Subjects were instructed to change lane as soon as they could recognize one of the symbols above.

can be made for the head-mounted display used in this study, resulting in the theoretical limit of 23.0 m,

$$Angle = \frac{2 \text{ pixels}}{42.7 \text{ pixels/}^\circ} = 0.04684^\circ = 2.81 \text{ arcmin}, \quad (2)$$

$$Distance = \frac{58 \text{ mm}}{\tan(2.81 \text{ arcmin})} = 71.0 \text{ m} \quad (3)$$

The distance to the road sign when the subjects start the maneuver to pass the sign was used as the measurement for this task. The hypothesis is that the lower visual acuity in the head-mounted display will make the subjects perform the lane change closer to the sign compared to when driving with projector-based graphics.

4.2.5 Narrow Curve Maneuver. The virtual test track contains a hairpin curve at each end to evaluate how well tight curves in urban environments are handled. The subjects were instructed to make a smooth turn with the vehicle centered in the available lane. The average deviation from an ideal path in the center of the lane was measured, as well as the number of steering wheel reversals needed to maneuver through each curve. The hypothesis is that the subject would benefit from having the added depth cues in the head-mounted display and the ability to turn around to see the full environment. These benefits should enable the subjects to drive the curve with smaller deviations and fewer steering wheel reversals.

4.3 Subjects

Twenty-five subjects (13 men and 12 women, mean age 40 years, range 21–65 years) were recruited from the general public via an open recruitment form on the institute's website. Subjects were required to have a valid driver's license and to be able to drive without using glasses, although wearing contact lenses was permitted. Glasses were eliminated due to space constraints inside the head-mounted display. Individuals who self-reported using medication that affected their balance were excluded from participation. Women who self-reported being pregnant were also excluded. All signed an informed consent form before participating in the study. All subjects in the study received two cinema tickets as a reward for their participation.

4.4 Procedure

Before the subjects drove the simulator, they read a written instruction of all the driving tasks they would be asked to perform. The test leader then administered a pre-drive **simulator sickness questionnaire (SSQ)** [18] to record a pre-test baseline. The subjects were then accompanied to the vehicle in the simulator and shown how to adjust the seat and cabin temperature for optimal comfort. When the subjects used a head-mounted

display, they were instructed how to adjust the interpupillary distance setting to their individual preference before putting it on. They were also shown how to adjust the headbands for a firm and comfortable fit.

Both conditions started with one lap around the virtual test track to familiarize the subjects with the simulator. During this lap, they were instructed to try out different speeds and changing lanes. The average time for completion of this familiarization sequence was 2 min 21.3 s (standard deviation 17.6 s). After the familiarization lap, they were instructed to stop before starting the real driving task. Before each task, the test leader repeated the instructions of the given task to the subject.

Once the subjects had completed all driving tasks of the first test condition, they were escorted from the simulator. The subjects responded to a post-test simulator sickness questionnaire as well as a questionnaire regarding perceived realism. This realism questionnaire used a Likert-scale with seven steps where the subjects were asked to rate the realism for factors such as longitudinal and lateral motion, as well as the visual and auditorial environment. Once the questionnaires were answered, the subjects were allowed to rest for five minutes before being accompanied back to the simulator to drive the second condition.

The average time to complete all driving tasks for a given display type condition was 10 min 26.1 s (standard deviation 1 min 6.6 s).

When the subjects had completed the second condition, they performed another post-test simulator sickness questionnaire and a realism questionnaire.

5 RESULTS

Three persons (one man and two women) had to abort the study due to severe motion sickness. One subject failed to perform most of the maneuvers as instructed, resulting in unusable data for this subject. Thus this subject is excluded from the analysis as well. Consequently, the analysis of driving behavior does not contain data from these four individuals.

5.1 Analysis of Driving Behavior

Analysis of variance was used to investigate the effects of the visual display types, the variation between subjects, and the interaction between these two factors. The *visual display type* was used as a fixed factor, while the *subject* was used as a random factor. Since the study used a within-group design, the conditions were performed in balanced order, and the *order number* was used as a fixed factor. In the speed perception analysis, the *reference speed* was used as a fixed factor.

A model selection was made for each of the response variables. *Visual display type* and *subject* was forced to be in the model even if they were not significant, while *order number*, *reference speed* (if present), and all interactions were allowed to be removed in the model selection. Each factor and each two-factor interaction were included in a preliminary model. Non-significant interactions were then removed. Non-significant main variables other than those forced to be in the model were removed unless they were part of a significant interaction.

The results of the analysis of visual display types, the difference between projectors and head-mounted displays are presented in Table 1 and discussed in detail in the following sections.

Significant variation between subjects can be seen for all measurements, except for the average deviation in the low-speed lane change task and the sign detection task (see Table 2).

The interaction effect between the *visual display type* (projectors vs. head-mounted display) and *subject* also revealed that different subjects react differently to changing between projectors and head-mounted displays (see Table 3), except for the steering wheel reversals in the medium speed lane change task.

5.1.1 Speed Perception. There is a significant difference in the speed perception between projectors and head-mounted display (see Figure 5 and Table 1). When the subjects drive using the projector-based setup, they tend to drive 3.247 km/h slower (standard error 1.199 km/h) compared to the reference speed. Whereas in the

Table 1. The Main Effect, i.e., Difference between Projector-based Graphics and Head-mounted Displays

Task	<i>p</i> value	Projectors		Head-mounted display		Difference	
		<i>Avg.</i>	<i>Std. Err.</i>	<i>Avg.</i>	<i>Std. Err.</i>	<i>Value</i>	<i>Std. Err.</i>
Speed perception	0.000	-3.247	1.199	2.480	1.199	-5.728	0.929
Low-Speed Lane change							
— <i>Average Deviation</i>	0.211	0.283	0.030	0.333	0.030	-0.051	0.039
— <i>Steering Wheel Reversals</i>	0.835	27.238	1.771	26.830	1.776	0.408	1.939
Medium Speed Lane change							
— <i>Average Deviation</i>	0.147	0.355	0.037	0.417	0.037	-0.062	0.041
— <i>Steering Wheel Reversals</i>	0.863	15.571	0.970	15.690	0.970	-0.119	0.685
Sign detection	0.000	47.859	1.659	13.025	1.693	34.834	2.102
Narrow curve maneuver							
— <i>Average Deviation</i>	0.182	0.601	0.070	0.662	0.070	-0.061	0.044
— <i>Steering Wheel Reversals</i>	0.000	8.695	0.593	5.619	0.593	3.076	0.586

head-mounted display setup, they drive 2.480 km/h faster (standard error 1.199 km/h). However, as can be seen in Table 2, the interpersonal differences are quite substantial, and the interaction effect between the visual display type and the subjects (see Table 3) shows that the difference between projectors and head-mounted display varies between subjects.

An ordering effect was observed for this task. The deviation from the reference speed was higher for the second display type ($p = 0.045$). Subjects who started driving with projectors had an average deviation of -2.376 km/h, i.e., slower than requested. The subjects who drove with the projectors as their second display type had a larger average deviation -4.104 km/h. The opposite was observed for the head-mounted display. The subjects who started driving with the head-mounted display drove on average 2.016 km/h faster than the requested speed. Those who had the head-mounted display as their second condition deviated more, with an average of 2.952 km/h.

When analyzing each reference speed separately, the head-mounted display condition approaches the reference speed when trying to match 40 km/h, but the magnitude of the absolute difference between the two display types is still similar (see Table 4).

5.1.2 Average Lateral Deviation. None of the tasks showed statistically significant effects at the 5% level for the average deviation. However, as shown in Figure 6, the average deviation exhibits a small tendency to increase when using a head-mounted display.

5.1.3 Steering Wheel Reversals. No change was found in steering wheel reversal behavior in the lane changing tasks, but the drivers use significantly fewer steering wheel corrections in the narrow curve when using a head-mounted display (see Figure 7).

5.1.4 Sign Detection. As expected, the subjects could detect the signs much further away using projectors compared to the head-mounted display (see Figure 8 and Table 1). When using projectors, they discern the symbol at a distance of 47.859 m (standard deviation 1.659 m). For the head-mounted display, the sign can be recognized at 13.025 m (standard deviation 1.693 m). The mean distances for both projectors and the head-mounted display are closer than the previously calculated theoretical maximum distance, but the theoretical ratio and the actual ratio are congruent.

Table 2. Intersubject Variation

Task	<i>p</i> value	Std. Dev.
Speed perception	0.000	4.588
Low-Speed Lane change		
— <i>Average Deviation</i>	0.241	—
— <i>Steering Wheel Reversals</i>	0.036	5.033
Medium Speed Lane change		
— <i>Average Deviation</i>	0.040	0.105
— <i>Steering Wheel Reversals</i>	0.000	3.857
Sign detection	0.287	—
Narrow curve maneuver		
— <i>Average Deviation</i>	0.000	0.284
— <i>Steering Wheel Reversals</i>	0.001	2.134

Table 3. Interaction Effect between the Visual Display Type and the Subject

Task	<i>p</i> value	Std. Dev.
Speed perception	0.010	2.143
Low-Speed Lane change		
— <i>Average Deviation</i>	0.000	0.118
— <i>Steering Wheel Reversals</i>	0.000	5.390
Medium Speed Lane change		
— <i>Average Deviation</i>	0.000	0.119
— <i>Steering Wheel Reversals</i>	0.070	—
Sign detection	0.000	6.281
Narrow curve maneuver		
— <i>Average Deviation</i>	0.000	0.125
— <i>Steering Wheel Reversals</i>	0.002	1.405

Note: The standard deviation describes how half the difference between the visual display types varies from subject to subject.

Table 4. Average Deviation in the Speed Perception Task Per Reference Speed

Reference speed	Projectors	Head-mounted display	<i>Difference</i>
20 km/h	−1.290	4.713	−6.003
40 km/h	−5.202	0.245	−5.447

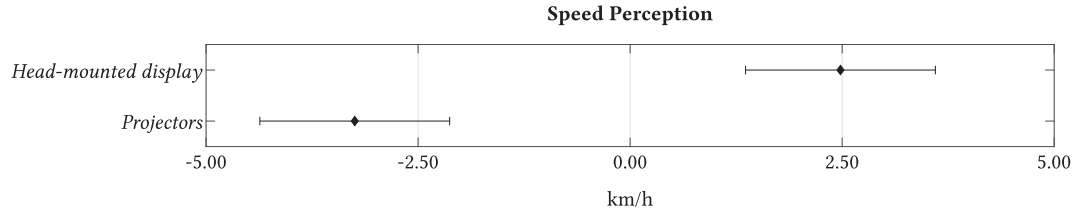


Fig. 5. Difference in speed perception for the visual display types.

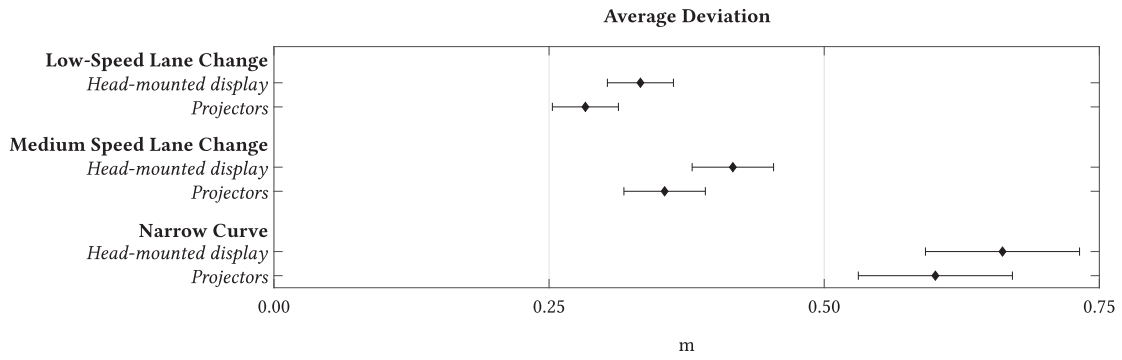


Fig. 6. Average deviation in the low-speed lane change, medium speed lane change, and in the narrow curve.

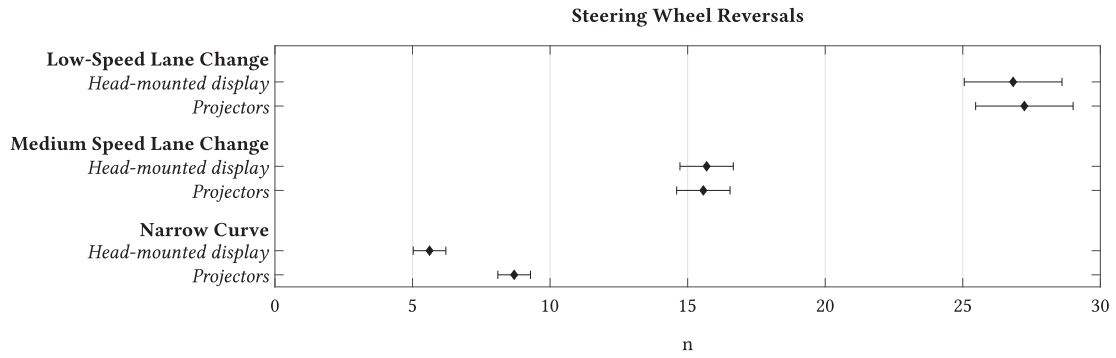


Fig. 7. Steering wheel reversals in the low-speed lane change, medium speed lane change, and in the narrow curve.

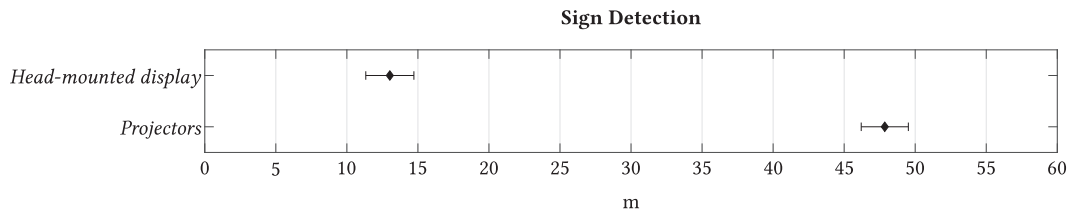


Fig. 8. Sign detection distance for the visual display types.

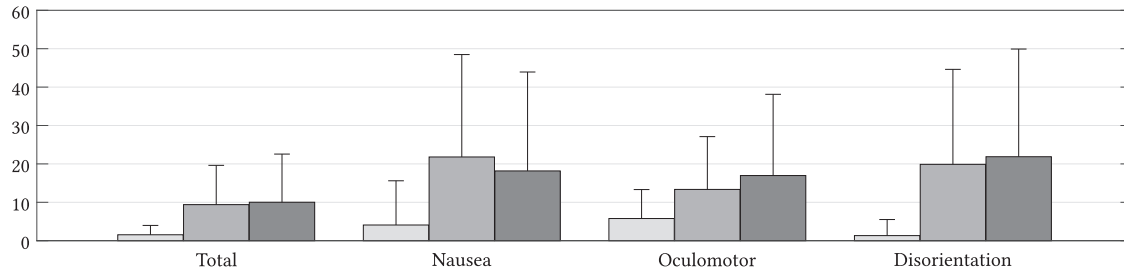


Fig. 9. The mean SSQ scores with error bars for one standard deviation. Bars ordered by pre-test, projector, and head-mounted display.

Table 5. The Simulator Sickness Scores for the Subjects Who Aborted after One Display Type Due to Severe Motion Sickness

Age	Gender	Display type	Total	Nausea	Oculomotor	Disorientation
25	Male	Projector	39.44	95.40	53.06	83.52
39	Female	HMD	21.96	28.62	30.32	55.68
59	Female	Projector	48.92	143.10	30.32	111.36

5.2 Simulator Sickness

Simulator sickness may accumulate over time, and the short break between the two conditions was probably not enough to eliminate such potential accumulation effects from the previous condition. However, the balanced design used in this study would reveal such ordering effects, and the analysis of the simulator sickness showed no significance for the ordering of the conditions.

The subjects who chose to abort due to severe motion sickness did so after driving only one display type condition. Therefore including them in the statistical analysis would skew the results in favor of the omitted display type. The recorded results in the simulator sickness questionnaire for the subjects who chose to abort are presented in Table 5.

Analysis of variance was used to study the effect of the visual display types and the variation between the remaining subjects on motion sickness and its sub-scales. The analysis showed no significant differences between the two visual display types for the resulting simulator sickness, neither for the total SSQ-score nor for any sub-scales (see Figure 9). The differences between individual subjects are more substantial than any difference between the display types.

5.2.1 Adaptation. An increase in simulator sickness can lead to behavioral adaptation as the subjects try to alleviate or minimize further increase. This adaptation can result in the subjects changing either their driving behavior, visual behavior, or both.

Analysis of the head movement of the individuals with the highest SSQ scores for the oculomotor subscale shows a clear difference in head-turning behavior compared to the group with the lowest SSQ scores (see Figure 10). This analysis suggests that one possible adaptation in behavior is a reduction in head-turning when subjects begin to experience oculomotor symptoms of motion sickness.

5.3 Perceived Realism

The analysis showed no statistically significant differences between the two display types in any posed questions in the realism questionnaire.

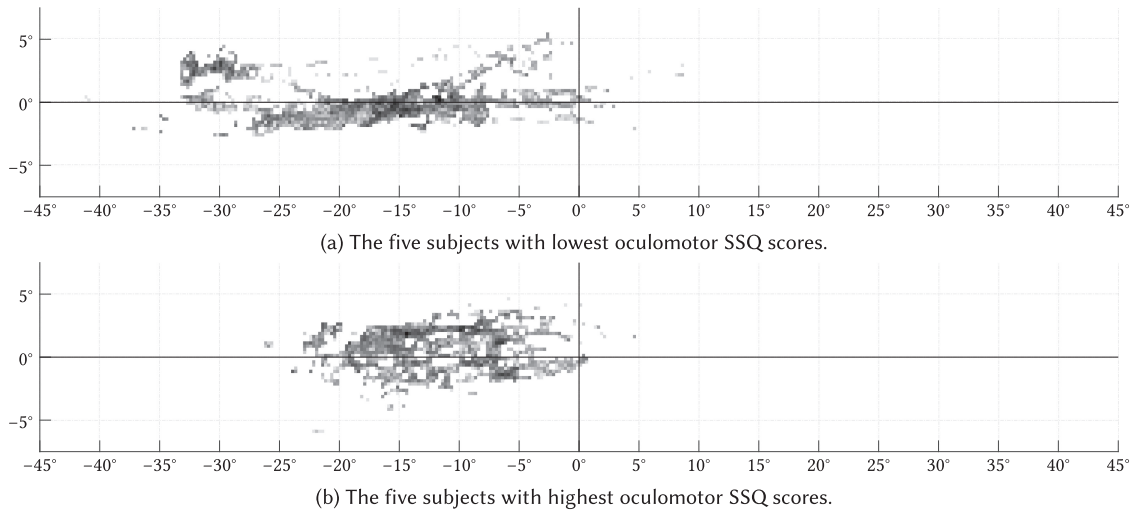


Fig. 10. Density plot of the head movement in the hairpin curve shows a limited head turn range for higher SSQ scoring individuals.

6 DISCUSSION

This study shows that it is possible to use a head-mounted display as the primary display in a dynamic driving simulator. The observed changes in lateral and longitudinal driver behavior are minor when comparing projectors and a head-mounted display.

The lateral driving behavior seems mostly unaffected by the use of a head-mounted display. A significant difference can be seen in the most extreme curve in the tested scenario, which might support the use of a head-mounted display for scenarios that involve tight curves, such as city driving. There is a non-significant trend in favor of projectors for the average lateral deviation, which would be consistent with previous research [13]. A possible explanation for this effect is a conflict with the *Vestibulo-ocular reflex* induced by system latency [27]. This reflex is triggered by head motion and makes the eyes rotate to keep the image of the environment fixed on the retina even though the head is moving. This reflex can cause conflicts when using a head-mounted display, since the reflex's compensatory actions introduce errors as the display is fixed relative to the head [16].

The longitudinal behavior also displays minor changes between the two studied display types. Previous research has shown that the perceived speed is higher than the reference speed when the horizontal field-of-view is large, and vice versa [34]. The available horizontal field-of-view in the head-mounted display used in this study is narrower than the field-of-view made available by the projectors. The users can rotate their heads to compensate for this difference. However, the speed perception task results suggest that the underestimation of speed in the head-mounted display may be a result of the lower field-of-view. Another factor that may impact these results is the lack of motion feedback in the longitudinal direction, although this affected both studied conditions.

The most substantial difference between the two imaging systems can be found in the sign detection task. This difference is not unexpected considering the technical specifications regarding available angular resolution. Nevertheless, special consideration must be taken to remedy the effect of the comparably low resolution in the current generation head-mounted displays. One possible way to remedy the low resolution in the head-mounted display is to scale up road signs to let drivers detect them at more realistic distances [37]. Although, just scaling up road signs might lead to unrealistic environments, which might break immersion. Scaling up other objects in the world would probably result in an even greater breaking of immersion. It could also lead to difficulties in estimating self-motion due to changes in the optical flow [14].

Sign detection is not the only task where high angular resolution is needed. Any type of target detection can benefit from having realistic sight distances, such as scenarios including overtaking, complex interaction with multiple actors, and detection of vulnerable road users. Thus an increase in angular resolution is highly desirable for the driving simulation use case.

In previous driving simulator research, the connection is unclear between introducing a head-mounted display and an increase in simulator sickness. However, most studies that report a large increase in simulator sickness were performed using static simulators [1, 8, 36, 37]. Regardless, there are studies performed in static simulators with non-significant differences as well [6]. For comparison, few studies explore the difference in a dynamic driving simulator. The early study by Schill et al. [32] showed no significant difference regarding simulator sickness. The study by Hartfiel and Stark [15] compared head-mounted displays in a dynamic simulator with the motion enabled and disabled. Enabling the motion in their case resulted in a small reduction in simulator sickness, but the effect was not significant. Similarly, this study could not identify any difference in simulator sickness between the two display types, which may be attributed to the motion system. Another reason might be that the scenario was relatively short, and a more prolonged scenario might have induced more simulator sickness.

6.1 Limitations and Future Work

The selected scenarios did not involve interaction with specific hardware inside the cabin, except for the steering wheel and pedals. Hence no hand tracking was used in this study, and the subjects could not see their own hands when they drove the simulator wearing the head-mounted display. Neither were any other representations of the driver's body present in the virtual world. This lack of self-embodiment was something that some of the subjects commented on during the post-drive questionnaire.

The study consisted of selected specific maneuvers, and it would be interesting to perform a similar study but instead use an extended sequence of more naturalistic driving.

It would also be desirable to investigate the effects of having a higher angular resolution, but that would be dependent on hardware evolution.

Another aspect that needs to be considered is the ergonomic effects of wearing a head-mounted display. The scenario used in this study is relatively short, but the added weight of the head-mounted display might be a considerable problem if the subjects were to drive a much longer scenario. Other ergonomic factors might also come into play during extended use; for example, the lenses may fog up due to increased temperature and humidity inside the head-mounted display.

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